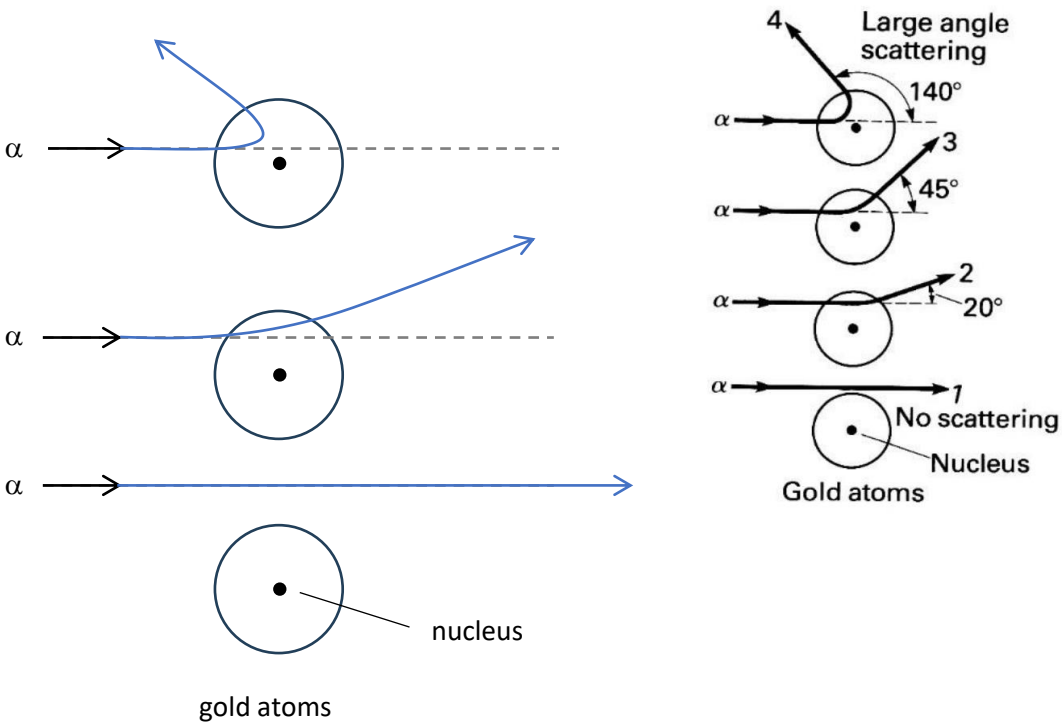


6(a)	 gold atoms	B3
(b)	<i>Experimental Evidence:</i> It was observed that most of the α-particles (more than 99%) will pass straight through or emerge scattered over a small angle. As α-particles are positively charged (and some α-particles are observed to be scattered / deflected), it shows the existence of a charged nucleus (protons are concentrated within the nucleus) within an atom. <i>Experimental Evidence:</i> Only a very small fraction of α-particles is observed to be backscattered (i.e. suffer deflections of more than 90°). This shows that the size of the nucleus is so small that the probability of an α-particle coming close enough to a nucleus to be deflected over a large angle is very low.	B1 A1 B1 A1
(c)(i)	The α -particle consists of 2 protons and 2 neutrons.	A1
(c)(ii)	Loss of mass, $\Delta m = (4.00260 + 9.01212) - (1.00867 + 12.00000)$ $= 0.00605 \text{ u}$ $= 1.0043 \times 10^{-29} \text{ kg}$ Energy equivalence of the loss mass, $E = (\Delta m)c^2$	B1 M1

	$= (1.0043 \times 10^{-29})(3.0 \times 10^8)^2$ $= 9.04 \times 10^{-13} \text{ J}$	A1
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(c)(iii)	Since energy is released after the reaction, the products have a higher total binding energy than the reactants. Note: As the total number of nucleons before and after the reaction remains the same, the product will have higher binding energy per nucleon (which means the product is more stable).	A1
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2023

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C2

Q7	Suggested Solutions	Marks
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